A Cooperative Approach to Humanoid Loco-Manipulation

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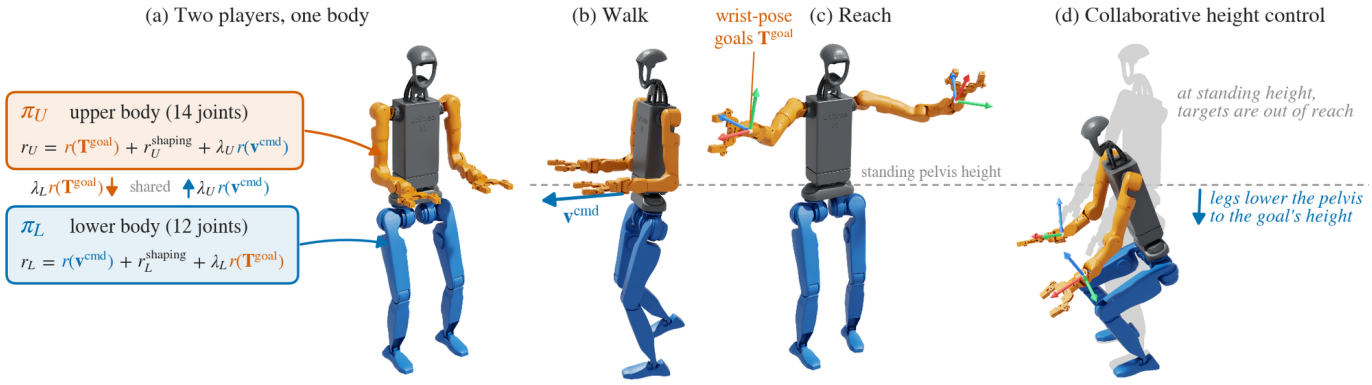


Fig. 1. Two agents cooperatively share one Unitree H1-2 humanoid. (a) An upper-body policy π_U tracks wrist-pose goals $T^{\text{goal}}_{\{\text{left, right}\}} \in SE(3)$ with 14 arm joints and a lower-body policy π_L tracks a velocity command $v^{\text{cmd}} \in \mathbb{R}^3$ with the 12 leg joints. (b) Walking is controlled solely by π_L. (c) During reaching, π_U tracks commanded wrist poses while π_L maintains balance. (d) For goals below standing reach, π_L exhibits an emergent height control behavior to extend the manipulation workspace.

***Abstract*—Whole-body loco-manipulation for humanoids is difficult due to high degree-of-freedom, complex dynamics with multiple objectives. In this paper we demonstrate how collaborative play between upper- and lower-body agents of a Unitree H1-2 humanoid can produce emergent whole-body behaviors like squatting and reaching, expanding effective workspace without motion priors or explicitly rewarding height-change or crouch behaviors. In sim-to-sim experiments in MuJoCo, the split-body policies place the wrists more accurately than a single whole-body policy (18.1–19.6 vs. 26.4 cm), and sharing more reward between the agents trades velocity tracking for fewer falls while reaching. The code for this project is available at https://github.com/maxconwa/Locomanipulation_game**

I. Introduction

Whole-body control (WBC) of bipedal humanoids remains of interest in a world built for humans. Where wheeled robots or mounted arms face topological and geographic constraints, humanoids will be able to integrate into human spaces like homes, hospitals, and manufacturing environments without the need for special accommodations.

However, there is significant disagreement on how best to proceed with producing robust whole-body control. Within the current paradigm of reinforcement learning for WBC there are two common broad approaches: a single, monolithic whole-body policy which balances multiple objectives such as motion imitation and balance, compared to separated lower- and upper-body policies responsible for balance and locomotion and manipulation and environment interaction, respectively. However, monolithic policies can suffer from reduced sample efficiency due to high-dimensional exploration and competing task objectives [1]. Furthermore, human motor control exhibits distinct upper-and lower-body control characteristics, including differences in force precision and the temporal organization of reaching and postural responses [2], [3].

Then, separated lower and upper bodies present a new question: how should these split policies coordinate with each other, during training and deployment? Prior literature has proposed training of split policies with critics which fully observe the whole-body [1], [4], [5]. More game-theoretic approaches have proposed adversarial play to ensure the lower-body is robust to the upper-body disturbances and the upper-body is robust to lower-body disturbances [6] or cooperative play to allow both policies to achieve their own goals independently with a shared reward [7]. In this work, we investigate how sharing task rewards between specialized upper- and lower-body policies affects whole-body coordination. We find that reward sharing induces emergent goal-dependent posture adaptation during reaching, such as the lower-body policy adjusting its height to assist the upper-body policy reach a faraway target, but also reveals a trade-off between manipulation performance and locomotion stability as the degree of cooperation varies.

The contributions of this work are twofold:

- Demonstration of emergent, goal-dependent posture adaptation through cooperative reward sharing, with char-

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acterization of reaching performance and stability.
- Comparison of cooperative split-body and monolithic whole-body policies, showing improved sample efficiency and reach accuracy, with sim-to-sim transfer.

II. Related Work

Split-body humanoid controllers co-train an upper- and a lower-body policy. ALMI [6] trains them adversarially through their commands: the lower body is trained against the upper-body motions that most often cause falls, and the upper body against a growing range of velocity commands. The policies become robust to each other, but because the lower body learns to resist the upper body, they never coordinate into whole-body behaviors such as squat and reach. FALCON [1] and HAFO [5] train the two agents cooperatively, making the lower body robust to external forces from manipulation or rope supports, but do not study how the two policies coordinate and require explicit height commands to achieve faraway reach targets. RoboDuet [7] jointly trains locomotion and manipulation policies for a quadruped, coupling arm-generated body-orientation commands and shared end-effector tracking rewards. We instead couple humanoid upper- and lower-body policies through tunable, bidirectional reward sharing, with no posture commands passed between them. SA2RT [8] takes a different game-theoretic approach, training one whole-body policy against an adversary that attacks its most vulnerable states, but as a single policy still trades locomotion against manipulation within one sum of weighted rewards.

III. Method

We formulate humanoid loco-manipulation as a two-player Partially Observable Markov Game (POMG), in which two agents control disjoint sets of actuators while sharing the same robot dynamics (Fig. 1). The upper-body policy π_U controls the 14 arm joints to track wrist-pose goals $T^{\text{goal}}_{\{l,r\}} \in SE(3)$, while the lower-body policy π_L controls the 12 leg joints to track velocity commands $v^{\text{cmd}} \in \mathbb{R}^3$ and maintain balance. Each policy acts on its own observation (Table III). Both are trained with MAPPO [9], and their centralized critics additionally the pelvis velocity and height, and the true wrist goals and poses. Although their actuation and task objectives are separated, the agents remain physically coupled: lower-body posture affects upper-body reachability, while arm motions affect whole-body stability. This raises a coordination problem: how should each agent account for the effect of its actions on the other's task? We investigate cross-agent reward sharing as a simple mechanism for inducing this cooperation without explicitly commanding coordinated whole-body motions.

A. Tasks

Training alternates between two task modes, indicated by $M \in \{0, 1\}$, which both policies observe. During navigation ($M = 0$), π_L tracks a velocity command $v^{\text{cmd}} = (v_x, v_y, \dot{\theta})$, with yaw commanded as a rate rather than a heading. Commands are drawn uniformly from $[-0.7, 0.7] \times [-0.3, 0.3]$ m/s and $[-0.5, 0.5]$ rad/s. Of these draws, 5% are zero (standing); of the remaining moving draws, 20% are replaced by pure turns in place ($|\dot{\theta}| \in [0.2, 0.5]$ rad/s) and 20% by pure lateral walks ($|v_y| \in [0.15, 0.3]$ m/s), each with a random sign. A navigation stretch lasts 4–8 s, and the command is resampled every 2.5–10 s. During navigation, T^{goal} is the arms' rest pose. During an arm goal ($M = 1$), which lasts 4 s, $v^{\text{cmd}} = 0$ and both wrist goals are held fixed in the world. Arm goals are sampled as described in Sec. III-B0a, and the mix of tasks follows the curriculum in Sec. III-D.

B. Rewards

Both task rewards use the kernel $\kappa_\sigma(x) = e^{-x^2/\sigma^2}$. The wrist-tracking reward is

$$r(T^{\text{goal}}) = \sum_{k\in\{l,r\}} \big[\, \kappa_{\sigma_c}(\|e_{p,k}\|) + \kappa_{\sigma_f}(\|e_{p,k}\|) + \kappa_{\sigma_\theta}(e_{\theta,k}) \,\big] + B\,\mathbb{1}^{\text{reach}}_t, \tag{1}$$

where $e_{p,k}$ and $e_{\theta,k}$ are wrist k's position and rotation errors to its true world-fixed goal. A coarse kernel ($\sigma_c = 0.25$ m) gives a gradient from far away and a fine kernel ($\sigma_f = 0.05$ m) rewards precise placement; $\sigma_\theta = 0.5$ rad. The bonus $B = 5$ is paid once, when both wrists first stay within 5 cm and 0.35 rad of their goals for 1 s. The velocity-tracking reward is

$$r(v^{\text{cmd}}) = 2\,\kappa_{\sigma_v}\big(\|v^{\text{cmd}}_{xy} - \bar{v}_{xy}\|\big) + 1.5\,\kappa_{\sigma_v}\big(\dot{\theta}^{\text{cmd}} - \bar{\dot{\theta}}\big), \tag{2}$$

where overbars denote pelvis-frame velocities averaged over one gait cycle (0.8 s), so that within-step sway is not rewarded as walking, and $\sigma_v = 0.25$ m/s for linear terms and rad/s for angualr terms.

We couple the agents' task objectives through a reward-sharing matrix C, while retaining independent shaping rewards:

$$\begin{bmatrix} r_U \\ r_L \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \lambda_U \\ \lambda_L & 1 \end{bmatrix}}_{C} \begin{bmatrix} r(T^{\text{goal}}) \\ r(v^{\text{cmd}}) \end{bmatrix} + \begin{bmatrix} r_U^{\text{shaping}} \\ r_L^{\text{shaping}} \end{bmatrix}. \tag{3}$$

Here, $\lambda_U, \lambda_L \in [0, 1]$ determine how much each agent values the other's task. At $\lambda_U = \lambda_L = 0$, the policies optimize independent task rewards; at $\lambda_U = \lambda_L = 1$, both receive the same combined task reward while retaining their individual shaping terms. Intermediate and asymmetric values vary the degree and direction of cooperation.

The key term is λ_L: sharing the wrist-tracking reward gives π_L an incentive to adjust its posture when doing so improves wrist tracking, even at some cost to its locomotion objective. In our reaching task, this lets the lower-body assist the upper-body without a height command, base height reward, or reference motion. The extent of this coordination is governed by the reward coupling rather than by a prescribed whole-body motion.

Each per-step reward is floored at zero before a fall penalty is added,

$$\tilde{r}_{i,t} = \max(0, r_{i,t}) - 5 \cdot \mathbb{1}^{\text{fall}}_t, \tag{4}$$

so that ending an episode early by falling is never preferable to continuing. Each agent maximizes its own discounted return $J_i = \mathbb{E}\big[\sum_t \gamma^t \tilde{r}_{i,t}\big]$.

a) Upper-Body Policy: $\pi_U : o_U \mapsto a_U \in \mathbb{R}^{14}$ drives the 14 arm joints. The torso joint belongs to neither policy and is held at its default angle, so only the legs can lower the shoulders. Wrist goals are drawn from a precomputed table of poses reachable while standing. Each goal is placed at the robot's position and heading, relative to the standing pelvis height, and then lowered by $z_{\text{offset}} \sim \mathcal{U}[0, z_{\max}]$, where $z_{\max}$ is set by the curriculum (Sec. III-D).

At each policy step, one damped-least-squares IK step [10], [11] moves each arm toward its goal, and π_U adds a learned residual:

$$q_U^{\text{tgt}} = q_U + \text{clip}(\Delta q^{\text{IK}}, \pm 0.1\,\text{rad}) + 0.2\,\bar{a}_U, \quad (5)$$

$$\Delta q_k^{\text{IK}} = J_k^\top \left(J_k J_k^\top + \mu^2 I\right)^{-1} e_k. \quad (6)$$

Here J_k is arm k's wrist Jacobian in the pelvis frame, e_k its 6-D pose error to the estimated goal (Sec. III-C), $\mu = 0.05$, and $\bar{a}_U$ the action low-passed at 3 Hz. The IK step provides a working reach from the first update, and the residual corrects for gravity, dynamics, joint limits, and self-collision. π_U does not observe the IK step.

b) Lower-Body Policy: $\pi_L : o_L \mapsto a_L \in \mathbb{R}^{12}$ sets the leg joint targets $q_L^{\text{tgt}} = q_{L,0} + 0.25\,a_L$ around the standing pose. It follows v^{cmd} and keeps the robot upright while the arms reach. Its shaping follows ALMI's gait and effort terms [6] (Table II). It does not observe the upper-body's actions or tracking errors; the wrist goals in o_L are its only direct cue that a goal is low.

C. Pelvis Estimator

Arm goals are fixed in the world, but the robot has no external localization, and the policies observe goals in the pelvis frame. We keep each pelvis-frame goal c_t on its world pose by dead reckoning, $c_{t+1} = \Delta_t^{-1} \circ c_t$, where Δ_t is the estimated pelvis motion over the step. The estimator affects only what the policies observe, since rewards always score the true goal. It is nevertheless what makes a change in height visible to the policies: without it, a pelvis-frame goal would descend with the pelvis, and crouching would never appear to bring the wrists closer. With it, the observed goal rises toward the wrists as the pelvis descends, consistent with the reward.

The estimator is an MLP (256, 128 units, ELU [12]). Its input is the last four steps of the sensor terms in o^{prop}, the 12 leg joint torques, and both policies' actions. Its output is the pelvis's mean linear and angular velocity over the step, which is integrated over the policy period $T_s = 0.02$ s to give Δ_t. It is trained online by mean-squared error against the simulator's true pelvis motion (Adam, learning rate 10^{-3}, every 24 steps) and frozen at evaluation.

D. Curriculum

Training starts with navigation only, until 80% of episodes end in a time-out rather than a fall. After that, each new task, on reset or when the previous task ends, is navigation or an arm goal with probability 0.5. Goal depth follows a per-environment level $k_d \in \{0, \ldots, 10\}$ that sets $z_{\max} = k_d/10$ meters. k_d rises when the wrists' mean closest approach over the environment's last five goals is within 12 cm, falls when it is beyond 18 cm, and drops by one after a fall during an arm goal. Judging closest approach rather than success lets k_d rise once a partial crouch brings the wrists near the goal, before the lower-body has fully learned to crouch.

TABLE I
RESULTS IN MUJOCO.

Metric	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	Whole-body
Reach avg. error (cm)	19.4	19.6	**18.1**	26.5
Pelvis rise, low → high (cm)	+1.3	+8.9	**+10.5**	+5.7
Wrists within 10 cm (%)	26.1	**37.0**	28.2	9.4
Vel. magnitude error (m/s)	**0.079**	0.083	0.299	0.309
Vel. direction error (°)	**5.9**	12.8	35.7	31.5
Falls / 1000 reach goals	12	11	**2**	12
Falls / 1000 navigation commands	1	**0**	12	28

IV. EXPERIMENTS

We train in Isaac Lab [13] with 4096 parallel environments on flat ground. Physics runs at 200 Hz, the policies at 50 Hz, and episodes last 20 s. Both policies are trained from scratch with MAPPO as implemented in skrl [14]. Each actor and critic is an MLP with hidden layers of 256, 128, and 64 units (ELU), and no parameters are shared. The actors are Gaussian with a state-independent standard deviation, bounded to $[0.05, 0.6]$ for π_L and $[0.05, 1.0]$ for π_U. Observations and critic states are normalized with running statistics, and the critics are fit to raw returns. Training runs for 90,000 steps per environment, or 3.7×10^8 transitions in 3,800 updates. This takes about 2.6 h on one NVIDIA RTX 4090. All experiments are run in MuJoCo to evaluate the learned policies outside of the physics simulator they are trained in.

To make the policies robust for sim-to-sim transfer, we randomize contact friction (static 0.6–1.2, dynamic 0.4–0.9), torso mass (-2 to $+4$ kg), velocity pushes of up to ± 0.4 m/s in x and y every 8–12 s, and observation noise (Table III). Table IV lists the MAPPO settings. Our experiments are designed to answer two questions:

1) How does λ impact the ability to reach and the tradeoff between the two tasks?
2) Is this split-body policy more sample efficient than a whole-body policy?

A. Reward Sharing Ablation

We train three pairs with $\lambda_U = \lambda_L \in \{0, 0.5, 1\}$ (Table I). Sharing barely changes the average reach error (18.1–19.6 cm), and all three pairs crouch for low goals, so the crouch itself does not require sharing. What sharing changes is how the crouch follows the goal: from the lowest to the highest goals the pelvis rises only 1.3 cm at $\lambda = 0$, which holds one crouch for every goal, but 8.9 and 10.5 cm at $\lambda = 0.5$ and 1. $\lambda = 0.5$ brings the wrists within 10 cm most often (37.0%) with nearly $\lambda = 0$'s speed error; $\lambda = 1$ falls least while reaching (2 per 1000 goals) but tracks velocity worst (0.299 m/s, 35.7°, 12 falls per 1000 commands). Rather than improving both tasks, λ sets the trade-off between them.

The crouch is a response to the goals π_L observes: when the $\lambda = 0.5$ lower body is given its navigation inputs ($M = 0$, wrist goals at the rest pose) while the arms reach, its median pelvis drop falls from 16 to 3 cm and falls rise from 11 to 345 per 1000 goals. At $\lambda = 0$, π_L also observes the goals but gains no reward from the wrist terms, producing a myopic crouch posture learned insensitive to goal height. Sharing also has a training cost: $\lambda = 0$, 0.5, and 1 pass depth level 9 after 43k, 45k, and 53k steps, mirroring walking speed error at the end of training (0.24, 0.28, and 0.30 m/s). Among the values we tried, $\lambda = 0.5$ balances the two tasks best.

B. Sample Efficiency

We train one policy $\pi_{wb} : o_{wb} \mapsto a_{wb} \in \mathbb{R}^{26}$ on the same task: it observes the union of o_L and o_U, and receives every term of r_L and r_U once. The pairs pass depth level 9 ($z_{\max} = 0.9$ m) after 43k–53k steps per environment (Fig. 3b); π_{wb} stays below level 1 until 50k steps and reaches level 9 only at 86k, 1.6–2.0× as many transitions despite 1.4× the parameters. In MuJoCo it has the largest reach error (26.5 cm) and most falls while navigating (28 per 1000), with velocity tracking no better than $\lambda = 1$'s.

The gap is already visible at the end of training in Isaac Lab. Over the last 2.4k steps, π_{wb} reaches 1.4% of arm goals, against 4.5–7.4% for the pairs, and 0.3% of the goals that need a crouch, against 2.3–2.5%. Its wrist orientation error stays at 0.72 rad, against 0.48–0.53 rad. In a deterministic rollout, it also fails at the lower body's task. Commanded to walk at 0.4 m/s, it steps in place while the $\lambda = 0.5$ pair walks at 0.32–0.36 m/s. On a goal 0.40 m below the rest pose, its wrists stop 6.9 cm from the goal, against 1.8 cm. A single policy must trade locomotion against manipulation within one reward and one advantage estimate over all 26 joints, while the split body gives each agent its own reward and critic. This is consistent with the faster, more even learning of the pairs, although we have not isolated which of these differences matters most.

V. Conclusion and Future Work

Our results suggest nuance in cooperation between specialized humanoid controllers. Even without reward sharing, the policies learn to crouch, and sharing task rewards helps improve posture response to a reaching goal. Stronger cooperation reduces falls during reaching but can compromise locomotion tracking, exposing a trade-off rather than a uniformly better controller. These observations raise a broader question for contact-rich loco-manipulation: how much should specialized controllers cooperate, and which controller should bear the cost of helping another? Our reward coupling is fixed and symmetric, but the appropriate degree of cooperation may change with the task, contact configuration, and required interaction forces. A more principled game-theoretic treatment could help characterize these competing objectives and inform how cooperative incentives should be structured and adapted. Investigating asymmetric and adaptive cooperation, particularly during physical manipulation and on hardware, is a natural next step, in addition to vastly improving the robustness and performance of the trained policies.

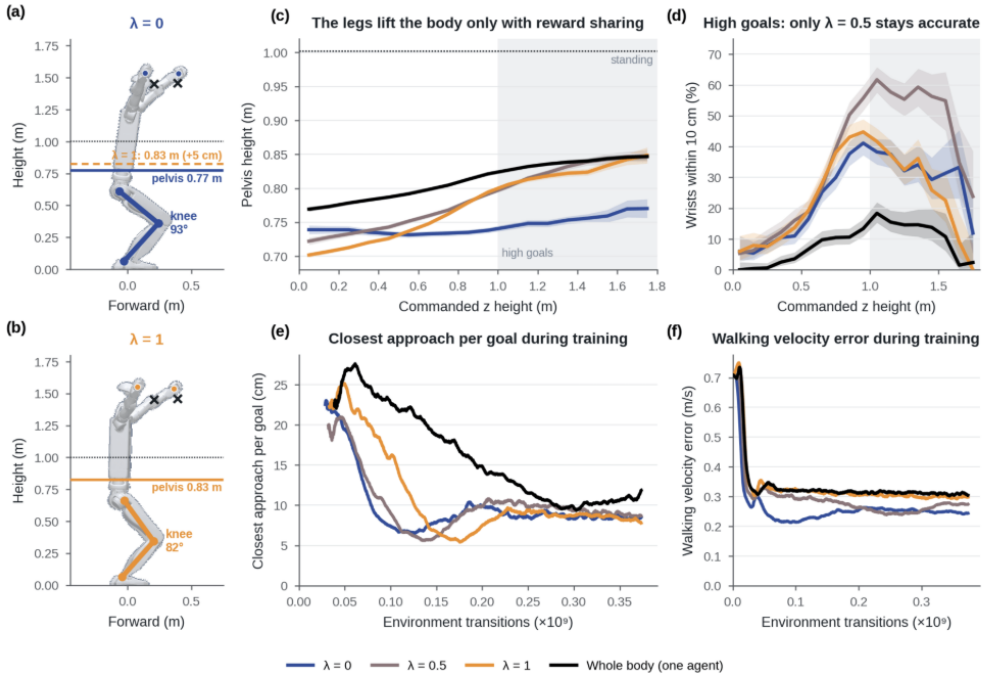


Fig. 2. Reward sharing lifts the body for high goals. (a, b) A representative high goal in MuJoCo. Without sharing ($\lambda = 0$) the knees stay bent (93° vs 82°) and the pelvis 5 cm lower. (c) Pelvis height and (d) wrists within 10 cm of target against commanded z height (3,168 goals per policy, 95% CI). Only $\lambda = 0.5$ stays accurate on high goals. The one-agent whole-body baseline stands tall but is least accurate. (e) Closest approach per goal and (f) walking velocity error during training.

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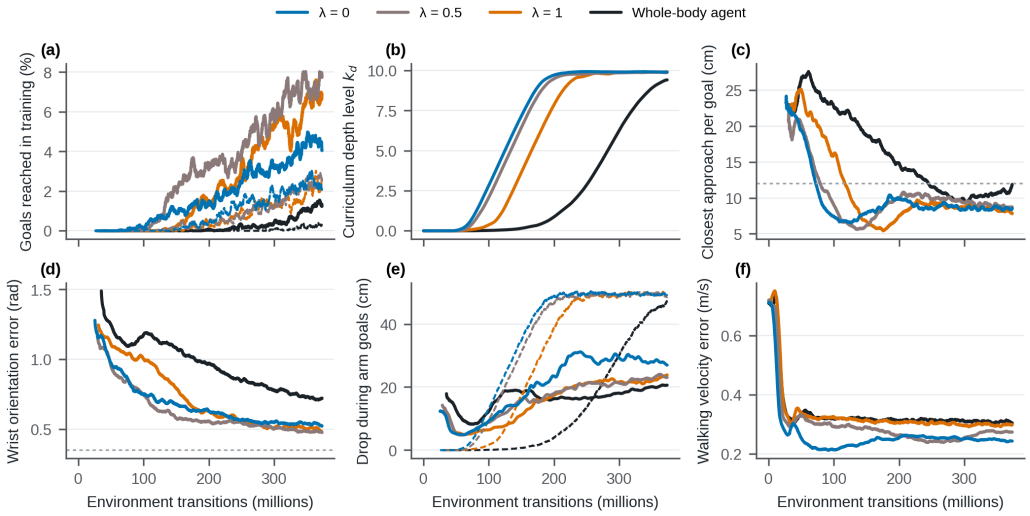
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APPENDIX



Dashed: (a) goals below standing reach, (e) the goals' lowering. Dotted: (c) 12 cm, the closest approach the depth level rises at; (d) 0.35 rad, the reach test's orientation tolerance. Smoothed over 50 updates.

Fig. 3. Sample efficiency of the two-agent policies ($\lambda = 0, 0.5, 1$) and the one-agent whole-body baseline, against environment transitions (policy steps $\times$ 4096 environments), smoothed over 50 updates. (a) Arm goals reached in training; dashed: goals below standing reach. (b) Depth curriculum level k_d. (c) Closest approach per goal; dotted: the 12 cm level-up threshold. (d) Wrist orientation error; dotted: the 0.35 rad reach tolerance. (e) Pelvis drop during arm goals; dashed: the goals' lowering. (f) Walking velocity error.

TABLE II

REWARD WEIGHTS PER SECOND (THE BONUS PER GOAL); TASK WEIGHTS FOLLOW EQ. (3), AND JOINT TERMS COUNT EACH POLICY'S OWN JOINTS. M: ARM-GOAL MODE; c_f, s_f: CONTACT AND SWING OF FOOT f (GAIT PERIOD 0.8 S, SWING FOR PHASE ≥ 0.55); $z^{ref} = 0.045 + 0.055\sin(\pi s)$ M; $\mathbb{1}_0$: $\|v^{cmd}\| < 0.1$; $(x)^+ = \max(0, x)$.

Term	Expression	r_L	r_U
Task $r(v^{cmd})$, Eq. (2)			
Linear velocity	$\kappa_{\sigma_v}(\|v_{xy}^{cmd} - \bar{v}_{xy}\|)$	2.0	$2.0\lambda_U$
Yaw rate	$\kappa_{\sigma_v}(\dot{\theta}^{cmd} - \bar{\dot{\theta}})$	1.5	$1.5\lambda_U$
Task $r(T^{goal})$, Eq. (1), per wrist			
Position, coarse	$\kappa_{\sigma_c}(\|e_p\|)$	λ_L	1.0
Position, fine	$\kappa_{\sigma_f}(\|e_p\|)$	λ_L	1.0
Orientation	$\kappa_{\sigma_\theta}(e_\theta)$	λ_L	1.0
Goal reached	once per goal	$5\lambda_L$	5
Shaping, both policies			
Alive	1	1.0	0.15
Joint torque	$\|\tau\|^2$	−1e−5	−1e−5
Joint velocity	$\|\dot{q}\|^2$	−1e−3	−1e−3
Joint acceleration	$\|\ddot{q}\|^2$	−2.5e−7	−2.5e−7
Action rate	$\|a_t - a_{t-1}\|^2$	−0.01	−0.01
Joint limits	excess past soft limits	−5.0	−5.0
Self-collision	pairs with an own link	−1.0	−1.0
Yaw rate in goals	$M\,\omega_z^2$	−1.0	−2.0
Shaping, lower body only			
Contact phase	$\sum_f \mathbb{1}[c_f \neq s_f]$	0.18	–
Swing clearance	$\sum_f s_f\,(z^{ref} - z_f)^+$	−25	–
Foot slip	$\sum_f c_f\,\|v_f\|^2$	−0.2	–
Feet / knee spacing	[0.3, 0.6] m / 0.3 m	1.0 / 0.2	–
Stance pelvis vel.	$\mathbb{1}_0\,\|v_{xy}\|^2$	−1.0	–
Vertical velocity	$\dot{z}^2$	−2.0	–
Roll/pitch rate	$\omega_x^2 + \omega_y^2$	−0.5	–
Pelvis tilt	$\|g_{xy}\|^2$	−0.25	–
Hip yaw/roll pose	$(1 - 0.8M)\,\|q_{hip} - q_{hip,0}\|^2$	−1.0	–
Ankle torque	$\|\tau_{ank}\|^2$	−5e−5	–
Ankle action rate	$\|\Delta a_{ank}\|^2$	−0.02	–
Foot contact force	excess over 1400 N	−0.01	–
Action beyond clip	excess past ±10 and limits	−0.02	–

TABLE III

POLICY OBSERVATIONS, IN THE PELVIS FRAME. BOTH POLICIES SHARE o^{prop} AND APPEND THEIR OWN TERMS. POSES ARE A POSITION AND A QUATERNION; NOISE IS UNIFORM, ± THE VALUE SHOWN.

	Term	Dim.	Noise
Shared, o^{prop}: 83			
$\ddot{p}^{imu}$	IMU linear acceleration	3	0.5 m/s^2
ω^{imu}	IMU angular velocity	3	0.2 rad/s
g	projected gravity	3	0.05
v^{cmd}	velocity command	3	–
M	mode (1: arm goal, 0: navigation)	1	–
$T_{l,r}^{goal}$	wrist goals	14	–
$q - q_0$	joint positions	27	0.01 rad
$\dot{q}$	joint velocities	27	1.5 rad/s
$\sin(2\pi\phi)$	gait clock, per leg	2	–
Lower body, $o_L = [o^{prop}, \cdot]$: 95			
$a_{L,t-1}$	previous action	12	–
Upper body, $o_U = [o^{prop}, \cdot]$: 123			
$\bar{a}_{U,t-1}$	previous filtered residual	14	–
$T_{l,r}^{wrist}$	wrist poses	14	0.005
$e_{l,r}$	wrist errors to goal	12	0.005

TABLE IV
MAPPO HYPERPARAMETERS, SHARED BY BOTH POLICIES UNLESS NOTED.

Parallel environments	4096
Rollout length	24 steps
Epochs $\times$ mini-batches	5×4
Discount γ / GAE λ_{GAE}	0.99 / 0.95
Clip ratio (policy, value)	0.2, 0.2
Learning rate	10^{-3}, KL-adaptive (target 0.02)
Gradient-norm clip	1.0
Entropy coefficient	0.01 (π_L), 0 (π_U)
Time-out bootstrapping	π_L only
Raw action clip	± 10
Actor / critic MLP	256, 128, 64 (ELU)
Action std bounds	$[0.05, 0.6]$ π_L, $[0.05, 1.0]$ π_U
Whole-body π_{wb}	512, 256, 128; KL target 0.04
Training steps per environment	91,200